Lesson 1: Powers, Exponent Vocabulary & the Product Law

Reviewing powers from Grade 9, then building the first exponent law.

1. What Is a Power?

An exponent counts repeated multiplication: $3^4 = 3 \times 3 \times 3 \times 3 = 81$. In the power t^k, t is the **base**, k is the **exponent**, and t^k is the **power**. The base can be a variable: $p \times p \times p = p^3$.

Example 1

State the base and the exponent, write each as a repeated multiplication, then evaluate where possible.

(a) 3^4 (b) 6^2 (c) x^5 (d) 8^3

2. Evaluating Powers

Expand, then multiply out. Watch two things: **order matters** ($2^5 = 32$ but $5^2 = 25$), and **brackets matter** with negatives — $(-3)^2 = 9$ but $-3^2 = -9$. With an odd exponent this distinction disappears: $(-2)^3 = -2^3 = -8$.

Example 2

Evaluate each power without a calculator.

(a) 2^5 (b) 5^2 (c) $(-3)^2$

(d) -3^2 (e) $(-2)^3$ (f) -2^3

3. Powers with Variable Bases

x^3 means $x \times x \times x$, and cannot be simplified until x is known. Substitute the *whole* value — sign included, in brackets — then evaluate: if $x = 4$, $x^3 = (4)^3 = 64$.

Example 3

Evaluate each power for the given value of the variable.

(a) a^3, $a = 2$ (b) b^2, $b = 5$ (c) c^4, $c = -2$ (d) d^3, $d = -1$

4. The Product Law

Expanding shows what happens when the bases match: $2^3 \times 2^2 = (2 \times 2 \times 2)(2 \times 2) = 2^5$, since $3 + 2 = 5$ factors — the exponents *add*. **Product Law:** $a^m \times a^n = a^{m+n}$ — keep the base, add the exponents. Bases must be identical; it does *not* apply to $2^3 \times 5^2$.

Example 4

Use the Product Law to simplify. Evaluate where possible.

(a) $3^2 \times 3^4$ (b) $6^3 \times 6^1$ (c) $m^2 \times m^5$ (d) $n^4 \times n \times n^2$

5. Applying the Product Law with Coefficients

A **coefficient** is the number out front (3 in $3x^2$). Multiply the coefficients, add the exponents: $3x^2 \times 5x^4 = (3 \times 5)(x^2 \times x^4) = 15x^6$.

Example 5

Use the Product Law to simplify each expression.
(a) $4x^2 \times 2x^3$ (b) $5a^3 \times 3a^2$ (c) $2y \times 6y^4$ (d) $7b^2 \times b^3$

Lesson 2: The Quotient Law & Power of a Power

Dividing powers subtracts exponents; raising a power to a power multiplies them.

1. The Quotient Law

Recall the Product Law: $a^m \times a^n = a^{m+n}$, so $4^2 \times 4^3 = 4^5 = 1024$. Division works the same way but with subtraction. Expanding, $2^5 \div 2^2 = \frac{2\times2\times2\times2\times2}{2\times2} = 2 \times 2 \times 2 = 2^3 = 8$, and $5 - 2 = 3$. **Quotient Law:** $a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$ (with $a \neq 0$) — keep the base, subtract the exponents. Bases must match.

Example 1

Use the Quotient Law to simplify.
(a) $7^5 \div 7^2$ (b) $\dfrac{(-3)^8}{(-3)^5}$ (c) $x^9 \div x^4$ (d) $\dfrac{y^{10}}{y^6}$

2. Quotient Law with Coefficients

When a coefficient sits in front, handle the coefficients and the powers **separately**: divide the coefficients normally, then subtract the exponents. For example, $8y^6 \div 2y^2 = (8 \div 2)(y^6 \div y^2) = 4y^4$.

Example 2

Simplify each expression.
(a) $12x^5 \div 4x^2$ (b) $\dfrac{20y^8}{5y^3}$ (c) $18a^9 \div 6a^4$

3. The Power of a Power Law

An outer exponent says how many copies of the inner power to multiply: $(2^3)^2 = 2^3 \times 2^3 = 2^6 = 64$, and $3 \times 2 = 6$. **Power of a Power Law:** $(a^m)^n = a^{mn}$ — keep the base, multiply the exponents. Check: $2^3 = 8$ and $8^2 = 64$.

Example 3

Use the Power of a Power Law to simplify.
(a) $\left(3^2\right)^3$ (b) $\left(2^3\right)^2$ (c) $\left(x^4\right)^3$ (d) $\left(y^2\right)^5$

Example 4

Use the Power of a Power Law to simplify.
(a) $\left(2^4\right)^2$ (b) $\left(6^2\right)^2$ (c) $\left(p^5\right)^3$ (d) $\left(q^3\right)^6$

4. Putting Both Laws Together

Some expressions need **both** laws. Simplify inside the brackets first by multiplying the exponents, *then* subtract using the Quotient Law.

Example 5

Use both the Quotient Law and the Power of a Power Law to simplify.

(a) $\dfrac{\left(x^3\right)^4}{x^5}$ (b) $\left(2^2\right)^3 \div 2^4$ (c) $\dfrac{\left(a^2\right)^5}{\left(a^3\right)^2}$

Lesson 3: Power of a Product and Power of a Quotient

Distributing an outer exponent onto every factor, then combining all five exponent laws.

1. Exponent Laws You Already Know

Three laws carry over from Lessons 1–2. **Product Law:** $a^m \times a^n = a^{m+n}$ — same base, add the exponents. **Quotient Law:** $a^m \div a^n = a^{m-n}$ — same base, subtract the exponents. **Power of a Power Law:** $(a^m)^n = a^{mn}$ — multiply the exponents.

Example 1

Warm up! Simplify using the Product Law, the Quotient Law, or the Power of a Power Law.

(a) $x^3 \times x^4$ (b) $y^7 \div y^2$ (c) $(m^2)^3$

2. The Power of a Product Law

Expanding shows what happens to a whole product: $(xy)^3 = (xy)(xy)(xy) = (x\,x\,x)(y\,y\,y) = x^3y^3$ — *each* factor picks up its own copy of the exponent. **Power of a Product Law:** $(ab)^n = a^n b^n$. Check with numbers: $(2 \times 3)^2 = 6^2 = 36$ and $2^2 \times 3^2 = 4 \times 9 = 36$.

Example 2

Use the Power of a Product Law to simplify each expression.

(a) $(3x)^2$ (b) $(2m)^4$ (c) $(-4y)^2$ (d) $(2ab)^3$

3. The Power of a Quotient Law

The same idea works on a fraction: $\left(\frac{a}{b}\right)^3 = \frac{a\,a\,a}{b\,b\,b} = \frac{a^3}{b^3}$. **Power of a Quotient Law:** $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$, where $b \neq 0$ — raise the numerator and the denominator to the power separately. Check: $\left(\frac{4}{5}\right)^2 = \frac{16}{25} = \frac{4^2}{5^2}$.

Example 3

Use the Power of a Quotient Law to simplify each expression.

(a) $\left(\dfrac{3}{4}\right)^2$ (b) $\left(\dfrac{x}{2}\right)^3$ (c) $\left(\dfrac{5}{y}\right)^2$ (d) $\left(\dfrac{m}{n}\right)^4$

4. Combining the Laws

Most expressions need more than one law. Strategy: first distribute the outer exponent with the Power of a Product or Power of a Quotient Law, then combine any matching bases with the Product or Quotient Law. For example, $(3x)^2 \times x^3 = 3^2x^2 \times x^3 = 9x^{2+3} = 9x^5$.

Example 4

Simplify each expression using two exponent laws.

(a) $(2x)^3 \times x^2$ (b) $(5m^2)^2$ (c) $(y^3)^2 \div y^4$ (d) $\left(\frac{a}{3}\right)^2 \times a^3$

5. Putting It All Together

These expressions combine three or more laws at once. Work step by step, applying one law at a time, and show every stage — a coefficient inside the brackets gets the outer exponent too.

Example 5

Simplify each expression completely.

(a) $(2x)^2(x^3)^2$ (b) $\dfrac{(3m^2)^2}{m^3}$ (c) $\left(\dfrac{2a^2}{b}\right)^3$

Lesson 4: Combining Exponent Laws

Choosing which law to use first, second and third in multi-step simplifications.

1. The Laws So Far, and the Order to Use Them

No new law here — just the five you already have: **Product** $a^m \times a^n = a^{m+n}$; **Quotient** $a^m \div a^n = a^{m-n}$; **Power of a Power** $(a^m)^n = a^{mn}$; **Power of a Product** $(ab)^n = a^n b^n$; **Power of a Quotient** $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$. Work the same checklist every time: **(1)** clear any bracket with an outer exponent (if the bracket holds a division, simplify inside it first), raising the coefficient *and* every variable to that power; **(2)** combine matching bases — add exponents when multiplying, subtract when dividing; **(3)** multiply or divide the coefficients last. For example, $(2x^3)^2 \times 3x^5$ becomes $4x^6 \times 3x^5$, then $x^{6+5} = x^{11}$ and $4 \times 3 = 12$, giving $12x^{11}$.

Example 1

Simplify $(3x^2)^3 \times 2x^4$, naming the checklist step you use at each stage.

Example 2

Simplify $\dfrac{(3x^4)^2}{x^3}$. Clear the bracket first, then use the quotient law.

2. Expressions with Two Variable Bases

When two different letters appear, treat each base separately: clear the bracket by raising *every* factor inside it to the outer power, then combine the x's with the x's and the y's with the y's. The coefficients are handled on their own at the end.

Example 3

Simplify. Each part uses two or three of the exponent laws.

(a) $(2a^3b)^2 \times 3a^2b^4$ (b) $\dfrac{(4x^2y^3)^2}{2xy^2}$ (c) $(x^2y^3)^3 \times (xy^2)^2$

3. Brackets That Contain a Division

If a bracket has a fraction inside it and a power outside it, simplify *inside* the bracket first with the quotient law, then apply the outer power to what is left. Doing the division first keeps the numbers small and prevents the most common error — forgetting to raise the coefficient.

Example 4

Simplify. Brackets first, then matching bases, then the coefficients.

(a) $\dfrac{(3x^2)^3 \times x^4}{9x^7}$ (b) $\left(\dfrac{6a^4b^2}{2a^2b}\right)^2 \times a^3$

Example 5

Simplify $\left(\dfrac{18x^6y^3}{6x^2y}\right)^2 \times x^3$. *Hint: which step handles the division inside the bracket, and which handles the power outside it?*

Lesson 5: Zero Exponents

Discovering why any nonzero base raised to the power zero is equal to one.

1. The Exponent Laws So Far

Lessons 1–4 gave five laws: **Product** $a^m \times a^n = a^{m+n}$, **Quotient** $a^m \div a^n = a^{m-n}$, **Power of a Power** $(a^m)^n = a^{mn}$, and **Power of a Product/Quotient** $(ab)^n = a^n b^n$ and $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$. Every one of those used a positive exponent, so nothing yet tells us what 3^0 means — multiplying 3 by itself 0 times is not an obvious idea the way $3^2 = 3 \times 3$ is.

2. Discovering the Rule: A Pattern

Drop the exponent by one and you divide by the base one more time: $3^4 = 81$, $3^3 = 27$ $(81 \div 3)$, $3^2 = 9$ $(27 \div 3)$, $3^1 = 3$ $(9 \div 3)$, so the next line must be $3^0 = 3 \div 3 = 1$. The same "divide by the base once more" pattern works for any nonzero base, since $a^1 = a$ divided by a is always 1 — the pattern predicts $a^0 = 1$.

3. Confirming It with the Quotient Law

Look at $5^4 \div 5^4$ two ways. By the Quotient Law it is $5^{4-4} = 5^0$; but it is also just a number divided by itself, which is 1. Both describe the same expression, so $5^0 = 1$. In general $a^n \div a^n = a^{n-n} = a^0$ and equals 1, giving the **Zero Exponent Law:** $a^0 = 1$ for any $a \neq 0$. The base must be nonzero, just as a denominator must be — 0^0 is undefined.

Example 1

Evaluate each power without a calculator.

(a) 4^0 (b) 9^0 (c) $(-6)^0$ (d) $(-1)^0$

4. Zero Exponents on Variables

The law works the same with a variable base: $x^0 = 1$ when $x \neq 0$. The danger is *what the exponent is attached to.* In $5x^0$ the 0 sits only on x, so $5x^0 = 5 \times 1 = 5$; in $(5x)^0$ the brackets make the whole product $5x$ the base, so $(5x)^0 = 1$. Same numbers, different answers — always check whether the exponent is on one factor or on an entire bracket.

Example 2

Simplify each expression.

(a) x^0 (b) $5x^0$ (c) $(5x)^0$ (d) $-2m^0$ (e) $(-2m)^0$

Example 3

Identify each statement as true or false.

(a) $7y^0 = 1$ (b) $(7y)^0 = 1$ (c) $x^0 = 0$ (d) $-4^0 = -1$

5. Combining the Zero Exponent Law with the Other Laws

Often the zero exponent is just one step inside a larger simplification. As soon as you spot a factor raised to the power 0, replace it with 1 — what remains is usually a straightforward product, quotient, or power-of-a-power question.

Example 4

Simplify each expression.

(a) $x^5 \times x^0$ (b) $\dfrac{a^6}{a^6}$ (c) $3b^0 \times 4b^2$ (d) $(2c^3)^0 \times 5c^4$

Example 5

Simplify each expression completely.

(a) $(2x^3y^0)^2 \times 3x$ (b) $\dfrac{(4a^2b^0)^2}{2a^3}$

Lesson 6: Negative Exponents

Extending the pattern past zero: a negative exponent means take the reciprocal.

1. Extending the Pattern Past Zero

From Lesson 5, $a^0 = 1$ for any nonzero base. Watch powers of 2 fall as the exponent drops by one, each line being the one above divided by 2: $2^3 = 8$, $2^2 = 4$, $2^1 = 2$, $2^0 = 1$. Keep dividing: $2^{-1} = 1/2 = 1/2^1$ and $2^{-2} = 1/4 = 1/2^2$ — the negative exponent gives the **reciprocal** of the matching positive power.

2. The Negative Exponent Rule

Negative Exponent Rule: $a^{-n} = 1/a^n$ for $a \neq 0$ — in words, "take the reciprocal and make the exponent positive." A negative exponent does *not* make the value negative: $2^{-3} = 1/2^3 = 1/8$, not -8. With a whole-number base bigger than 1, the answer is always a positive fraction less than 1. With a negative base, brackets matter just as in Lesson 1: $(-2)^{-2} = 1/(-2)^2 = 1/4$, but $-2^{-2} = -1/2^2 = -1/4$.

Example 1

Write each power with a positive exponent, then evaluate.

(a) 3^{-2} (b) 4^{-1} (c) 5^{-3} (d) 10^{-2}

3. Variable Bases

When the base is a variable there is no final number to compute — just apply the rule and stop: $x^{-3} = 1/x^3$. The same works with a letter exponent, so $a^{-n} = 1/a^n$.

Example 2

Write each expression with a positive exponent.

(a) x^{-3} (b) y^{-5} (c) m^{-1} (d) a^{-n}

4. Negative Exponents in the Denominator

If the negative exponent is already in a denominator, apply the rule twice: $1/a^{-n}$ becomes 1 divided by $1/a^n$, which is a^n. **Reverse Rule:** $1/a^{-n} = a^n$. So a negative exponent downstairs flips up to the numerator and turns positive — exactly the reverse of Section 2.

Example 3

Simplify each expression so the exponent is positive, then evaluate where possible.

(a) $\dfrac{1}{2^{-3}}$ (b) $\dfrac{1}{3^{-2}}$ (c) $\dfrac{1}{x^{-4}}$ (d) $\dfrac{1}{y^{-6}}$

5. Coefficients

A negative exponent applies only to the base written beside it, never to the coefficient out front. In $6x^{-3}$ the exponent -3 belongs to x alone, so $6x^{-3} = 6 \times 1/x^3 = 6/x^3$; the 6 stays upstairs.

Example 4

Simplify each expression, writing your answer with a positive exponent.

(a) $6x^{-3}$ (b) $2a^{-1}$ (c) $\dfrac{9}{y^{-2}}$ (d) $\dfrac{8}{b^{-4}}$

6. Negative Exponents from the Quotient Law

You will often meet a negative exponent as the *result* of the quotient law, whenever the exponent being subtracted is the larger one. Subtract as usual, then finish by rewriting the answer with a positive exponent.

Example 5

Simplify using the quotient law, then rewrite the result with a positive exponent.

(a) $x^2 \div x^5$ (b) $m^3 \div m^8$ (c) $\dfrac{12x^2}{3x^6}$ (d) $\dfrac{20x^5y^2}{4x^2y^6}$

Lesson 7: Combining Zero, Negative and Positive Exponents

No new rule — just every exponent law working together in one expression.

1. The Complete Set of Exponent Laws

All seven rules, in one place: $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $(ab)^n = a^n b^n$; $(a/b)^n = a^n/b^n$; $a^0 = 1$; and $a^{-n} = 1/a^n$ (the last two need $a \neq 0$). Nothing new is added this lesson — the laws are just combined. One habit matters throughout: **finish every answer with positive exponents only.**

Example 1

Simplify $4x^3 \cdot x^{-7} \cdot x^0$. Write the answer with positive exponents.

2. A Three-Step Checklist

Step 1: brackets first. If a bracket has a power outside it, raise the coefficient and every variable inside to that outer power. **Step 2: combine matching bases.** Add exponents where like bases multiply, subtract where they divide — coefficients and each variable separately. **Step 3: zero and negative exponents last.** Replace any leftover a^0 with 1 (a zero exponent can be cleared as soon as you see it), then flip any negative exponents using $a^{-n} = 1/a^n$.

Example 2

Simplify $12a^{-2} \div 4a^5$. Write the answer with positive exponents.

3. Brackets with Negative Outer Powers

A negative exponent outside a bracket still multiplies through: $(5x^3)^{-2} = 5^{-2}x^{-6}$, and $5^{-2} = 1/25$. Multiply the outer exponent by *each* inner exponent, signs included, and save the flip to positive exponents for the very last step.

Example 3

Simplify. Write every final answer with positive exponents.

(a) $(3x^2)^{-2} \times x^5$ (b) $\frac{(2a^3b)^2}{4a^7b^{-1}}$ (c) $(x^{-2}y^3)^{-3}$

4. Fractions Raised to a Power

Simplify inside the bracket before applying the outer power — it keeps the numbers small. A negative outer power on a fraction simply inverts it: $(a/b)^{-n} = (b/a)^n$. So $(2/x)^{-2} = (x/2)^2 = x^2/4$.

Example 4

Simplify. Write the final answer with positive exponents.

(a) $\left(\frac{2a^4}{a^{-2}}\right)^{-2}$ (b) $\frac{(3x^{-1}y^2)^2}{9x^{-4}y^5}$

5. Putting All Three Steps Together

Longer expressions are the same checklist, run once. A zero exponent hiding inside a bracket is cleared in Step 1 ($y^0 = 1$, so it disappears), and only after every base is combined do you decide which exponents still need flipping.

Example 5

Simplify $\dfrac{(2x^3y^0)^{-2} \times 6x^5}{3x^{-4}}$. Write the answer with positive exponents.

Lesson 8: Scientific Notation

Writing very large and very small numbers compactly using powers of ten.

1. Why Scientific Notation?

Some numbers are awkward to write out: the Earth–Moon distance is about 384 000 000 m, and a human hair is about 0.00007 m wide. It is easy to miscount zeros and hard to compare such numbers at a glance. Scientific notation fixes this: $384\,000\,000 = 3.84 \times 10^8$ and $0.00007 = 7 \times 10^{-5}$. Recall that each place value is a power of 10 ($10^3 = 1000$, $10^0 = 1$, $10^{-3} = 0.001$), so moving a digit one place to the left multiplies its value by 10; one place to the right multiplies it by 10^{-1}.

2. What Is Scientific Notation?

A number is in **scientific notation** when it has the form $a \times 10^n$, where n is an integer (positive, negative, or zero) and the coefficient satisfies $1 \le a < 10$ — that is, a has exactly one nonzero digit before its decimal point. So 4.2×10^5 and 9×10^0 are correct, but 52×10^3 is not ($52 \ge 10$) and 0.5×10^{-2} is not ($0.5 < 1$).

3. Standard Form → Scientific Notation

Place the decimal point right after the **first nonzero digit**, count how many places it moved, and use that count as the exponent. If the number is 10 or larger the decimal moves **left** and n is **positive**: $47\,000 \to 4.7$ is 4 places left, so $47\,000 = 4.7 \times 10^4$.

Example 1

Write each number in scientific notation.

(a) 6 000 (b) 52 000 (c) 730 000 (d) 8 400 000

For a number smaller than 1 the decimal moves **right** instead, which makes the exponent **negative**: $0.0004 \to 4$ is 4 places right, so $0.0004 = 4 \times 10^{-4}$. A number from 1 up to 10 needs no hops: $n = 0$, e.g. $5.9 = 5.9 \times 10^0$.

Example 2

Write each number in scientific notation.

(a) 0.003 (b) 0.00052 (c) 0.0000081 (d) 0.06

4. Scientific Notation → Standard Form

Reverse the process: move the decimal point of the coefficient $|n|$ places — **right** if n is positive, **left** if n is negative — filling in zeros as placeholders. For example 3.4×10^{-5} moves 5 places left to give 0.000034.

Example 3

Write each number in standard notation.

(a) 5×10^3 (b) 3.4×10^5 (c) 7×10^{-3} (d) 2.6×10^{-5}

Example 4

Convert each real-world fact into the form requested — some go each way.

(a) Earth's average distance from the Sun is about 150 000 000 km. Write this distance in scientific notation.

(b) A red blood cell is about 0.000007 m wide. Write this width in scientific notation.

(c) A bacterium was measured to be 3.5×10^{-6} m long. Write this length in standard notation.

(d) A tech company's total assets were 2.4×10^9 dollars. Write this amount in standard notation.

5. Comparing and Ordering

To compare two numbers in scientific notation, compare the **exponents** first — the smaller exponent gives the smaller number. Only when the exponents are equal do you compare the coefficients.

Example 5

Compare and order, justifying with the exponents.

(a) Order from smallest to largest: a bacterium (2×10^{-6} m), a human hair (7×10^{-5} m), and a grain of sand (5×10^{-4} m).

(b) City A has a population of 3.2×10^6 people and City B has 8×10^5 people. Which city has the greater population? Justify your answer using the exponents.

Lesson 9: Rational Exponents and Radical Form

Fractional exponents are radicals in disguise: converting both ways, then applying the laws.

1. Unit Fractional Exponents: What Does $a^{1/n}$ Mean?

The Product Law should keep working for fractions, so $5^{1/2} \times 5^{1/2} = 5^1 = 5$ — exactly what $\sqrt{5} \times \sqrt{5}$ does, forcing $5^{1/2} = \sqrt{5}$. Likewise $3^{1/3} \times 3^{1/3} \times 3^{1/3} = 3$, so $3^{1/3} = \sqrt[3]{3}$. In general, for $a \geq 0$, $a^{1/n} = \sqrt[n]{a}$: the **denominator names the root**. With an odd root (a cube root, a fifth root) the base can be negative: $(-8)^{1/3} = \sqrt[3]{-8} = -2$. For a fraction base, root the top and the bottom: $\left(\frac{16}{25}\right)^{1/2} = \frac{4}{5}$.

Example 1

Evaluate each power without using a calculator.

(a) $25^{1/2}$ (b) $64^{1/3}$ (c) $16^{1/4}$ (d) $100^{1/2}$

2. General Rational Exponents: $a^{m/n}$

The numerator is just an ordinary power, applied before or after the root: $a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$. Both orders agree — $8^{2/3} = (\sqrt[3]{8})^2 = 2^2 = 4$ and $8^{2/3} = \sqrt[3]{8^2} = \sqrt[3]{64} = 4$ — but rooting *first* keeps the numbers small, so take that path whenever the base has a nice root.

Example 2

Evaluate each power without a calculator. Take the root first to keep the numbers small.

(a) $9^{3/2}$ (b) $27^{2/3}$ (c) $16^{3/4}$ (d) $4^{5/2}$

3. Converting Between Radical Form and Rational Exponent Form

The same correspondence holds for variable bases, and it runs in *both* directions: $x^{m/n} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$, and any radical $\sqrt[n]{x^m}$ can be rewritten as $x^{m/n}$. The index becomes the denominator and the inside power becomes the numerator. A negative rational exponent still means "take the reciprocal," just as in Lesson 6.

Example 3

Write each power as an equivalent radical.

(a) $x^{1/5}$ (b) $y^{2/3}$ (c) $w^{3/4}$ (d) $z^{-1/2}$

Example 4

Write each radical as an equivalent power with a rational exponent.

(a) $\sqrt[3]{x^2}$ (b) $\sqrt[4]{y^3}$ (c) $\sqrt{z^5}$ (d) $\frac{1}{\sqrt[3]{w^2}}$

4. Applying Exponent Laws to Rational Exponents

Every law from this unit works unchanged on rational exponents — there is no need to convert to radicals first. For the Product Law, add the exponents over a common denominator: $a^{1/2} \times a^{1/3} = a^{\frac{3}{6}+\frac{2}{6}} = a^{5/6}$. Convert to radical form only at the very end, once the exponent is fully simplified.

Example 5

Simplify each expression. Leave your answer as a power with a positive exponent.

(a) $b^{1/4} \times b^{1/2}$ (b) $c^{5/6} \div c^{1/6}$ (c) $\left(d^{1/3}\right)^6$ (d) $\left(4g^{1/2}\right)^2$

Lesson 10: Unit Review: Exponents

Every law, zero and negative exponents, scientific notation and radicals — one last pass.

1. The Product Law

Multiplying powers with the **same base**: keep the base and **add** the exponents, $a^m \cdot a^n = a^{m+n}$. Multiply any coefficients separately.

Example 1

Evaluate or simplify.

(a) Evaluate $6^2 \times 6^3$. (b) Simplify $5t^4 \times 3t^2$.

2. Quotient Law and Power of a Power

Dividing powers with the same base: **subtract** the exponents, $a^m \div a^n = a^{m-n}$. A power raised to another power: **multiply** the exponents, $(a^m)^n = a^{mn}$.

Example 2

Simplify, using the quotient law, the power of a power law, or both.

(a) $k^{11} \div k^5$ (b) $(t^3)^4$ (c) $\dfrac{(g^2)^5}{g^4}$

3. Power of a Product and Power of a Quotient

Raise *each* factor to the outside power: $(ab)^n = a^n b^n$, and $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ — numerator and denominator both.

Example 3

Simplify each expression.

(a) $(4h)^2$ (b) $\left(\dfrac{r}{5}\right)^3$

4. Combining Exponent Laws

Clear any bracket carrying an outer exponent *first*, then combine matching bases with the product or quotient law, and handle the coefficients last.

Example 4

Simplify each expression.

(a) $(2t^3)^2 \times 5t^4$

(b) $\dfrac{(2w^3)^2}{4w^2}$

5. Zero Exponents

Any nonzero base to the exponent 0 equals 1: $a^0 = 1$. Check what the 0 is attached to — $8k^0$ raises only k, but $(8k)^0$ raises the whole product.

Example 5

Evaluate each pair, and explain why they differ.

(a) $8k^0$ and $(8k)^0$

(b) -9^0 and $(-9)^0$

6. Negative Exponents

A negative exponent means "reciprocal, then make the exponent positive": $a^{-n} = \frac{1}{a^n}$. It never makes the value negative, and it never moves a coefficient outside the power.

Example 6

Evaluate, or rewrite with a positive exponent.

(a) Evaluate 4^{-2}.

(b) Rewrite $9t^{-3}$.

(c) Evaluate $\dfrac{1}{5^{-2}}$.

7. Combining Zero, Negative, and Positive Exponents

Same checklist as before — brackets, then matching bases — with zero and negative exponents cleaned up last so the final answer carries positive exponents only.

Example 7

Simplify. Write each answer with a positive exponent.

(a) $\dfrac{15w^{-2}}{3w^4}$

(b) $(2t^{-3})^2 \times t^5$

8. Scientific Notation

Scientific notation is $a \times 10^n$ with $1 \le a < 10$ and n an integer. To multiply, multiply the leading factors and add the exponents of 10, then adjust a back into range.

Example 8

Answer each question, giving results in proper scientific notation where asked.

(a) Evaluate $(6.2 \times 10^3) \times (4.0 \times 10^{-7})$, and write the answer in proper scientific notation.

(b) A grain of pollen has a diameter of about 2.5×10^{-5} m. About how many grains, laid end to end, would span a length of 0.5 m?

9. Rational Exponents and Radical Form

A fractional exponent is a root and a power together: $a^{m/n} = \sqrt[n]{a^m}$ — denominator is the index, numerator is the power. A negative rational exponent still means reciprocal.

Example 9

Answer each question.

(a) Express $t^{3/8}$ in radical form.

(b) If $p = 16$ and $q = -8$, evaluate $p^{3/2} - q^{-2/3}$.

10. Mixed Review: Putting It All Together

Each question below combines more than one skill from the unit — exactly the kind of question that shows up on the unit test.

Challenge 1

Simplify $\dfrac{(2t^3)^{-2} \times 8t^5}{4t^{-3}}$, expressing your answer with a positive exponent.

Challenge 2

Simplify $s^{5/6} \times s^{1/6}$. Then, if $s = 4\,000\,000$, express the numeric value of the result in scientific notation.

Challenge 3

The area of a square garden bed is $(3 \times 10^2)^2$ cm^2.

(a) Express the area in proper scientific notation.

(b) The bed is divided into square tiles, each with an area of 9×10^2 cm^2. How many tiles cover the bed?